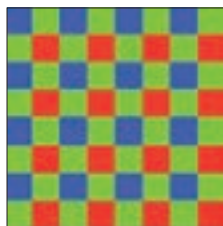


an RGB value for each site. This image is then compiled and displayed on the LCD screen.

The more the pixels on your sensor, the higher the resolution of the image sensor and, thus, the more detailed is the picture. To put the whole pixel thing into perspective, a 6 megapixel camera would give you the same resolution as the best 35 mm film cameras and anything above 8 megapixels will produce pictures that professionals dream of.

Sensors and Colour

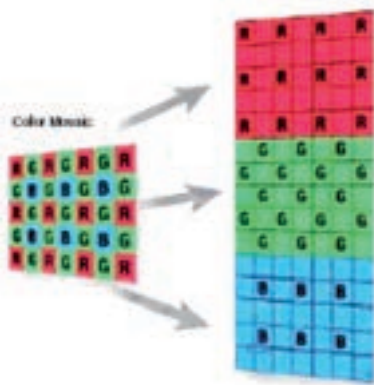
Sensors only register black and white. To fill the image with colour, it the sensors use a layer of colour filters called a 'colour filter array' (CFA) – either Bayer or Foveon, but usually Bayer when it comes to your typical digital camera. Typically, cameras use separate RGB (Red, Green and Blue, for the uninitiated) colour filter arrays, but some use the CYGM (Cyan, Yellow, Green and Magenta) array. The Bayer filter pattern consists of rows of red and green filters separated by a row of blue and green filters.



Chequered flag filters

Digital cameras most commonly use a 'Bayer mask' over the sensor, which lets the processor figure out the colour. Each square is made of four pixels, one red, one blue and two

green – more greens because our eyes pick up green better than blue or red. The best cameras use three different sensors – each with a different filter, using a 'beam splitter' to deflect the light to the sensors. The same beam of light hits each sensor, but, on account of the filters, each registers a different colour. This, however, needs more space and makes the camera a little



Filters that scatter